

Course Code : ECT 359-4

QTSR/RW – 24 / 1671

**Sixth Semester B. Tech.(Electronics and Communication Engineering)
Examination**

INTRODUCTION TO INTERNET OF THINGS

Time : 3 Hours]

[Max. Marks : 60

Instructions to Candidates :—

- (1) All questions are compulsory and carry marks as indicated.
 - (2) Due credit will be given to neatness.
 - (3) Assume suitable data and illustrate answers with neat sketches wherever necessary.
-
1. (a) For a Use Case, recommend an architecture with details and justification. 6(CO1,5)
(b) As an IoT Application Developer, select an Access Technology for Baby Monitoring application. Justify. 4(CO1,5)
 2. (a) Compare Raspberry pi and ESP32 board. 3(CO1)
(b) Describe the steps to enable Spl interface on Rpi using terminal and through GUI. 3(CO2)
 3. (a) Write a bash script to read name and address from a file “ employee_data.CSV” and display the line using loop with the following message “Hello< from the file >, How are you ? 6(CO2)
(b) Write a python script to execute a python source file “video_demo.py” with two input arguments stored in an arrays of length 8 and 3, respectively. 6(CO1,3,4)
 4. (a) Write a python script to perform convolution of an array X= [1, 2, 3, 4, 5, 6, 7, 8, 9, 10] and array Y = [-1, 0, 1] 5(CO1,3,4)

QTSR/RW-24 / 1671

Contd.

- (b) Write a python script to send an alert to a Thingsboard server whenever a PIR sensor detects motion at home and frequency of alert should be once in a minute. 5(CO1,3,4)
5. (a) Write a python script to read an image frame from USB camera and detect red color ball is present or not. 5(CO1,3,4)
- (b) Write a python script to create a flask server for face detection and sending an email with image of face detected. 6(CO4)
6. (a) Identify the challenges of implementing an IoT Use case for Computer Based Patient Diagnosis System for cancer in India How would you overcome those? Justify. 7(CO1,5)
- (b) Suggest and explain a protocol to monitor storms in a ocean environment using sensors with power constraints. 4(CO1)

